

ONeRec: Towards Openness-Aware and Adaptive Proactive News Recommendation

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Abstract

Proactive news recommendation seeks to guide users over extended interaction sessions towards a cultivated interest in targeted news, thereby shaping public opinion and contributing to social stability. Conventional news recommendation algorithms, by contrast, are largely passive: they rely solely on a user's historical preferences, a practice that exacerbates filter-bubble effects and opinion polarization. To mitigate these drawbacks, proactive news recommendation strategically adjusts the sequence of suggested articles so that users gradually cultivate an interest in a target. This paradigm, however, presents three central challenges: (i) accurately modeling a user's receptiveness to novelty; (ii) tracking evolving interests across multiple rounds of proactive recommendation; and (iii) selecting intermediary articles that balance immediate relevance with long-term target guidance. To tackle these challenges, we introduce **ONeRec**, a novel framework towards user **O**penness-aware and adaptive proactive **N**ews **R**ecommendation. ONeRec steers users towards target news by adaptively recommending target-relevant intermediate news items according to the user's openness and current interest. ONeRec incorporates two personalized mechanisms: an openness coefficient, derived from reading history, that models a user's tolerance for novelty and balances interest matching with target guidance; and an evolutionary coefficient, which dynamically updates user interest as they engage with recommended news. To support offline training and evaluation, we further employ a Large Language Model agent to simulate user feedback. Extensive experiments on the public MIND dataset demonstrate that **ONeRec** consistently outperforms strong baselines in proactive news recommendation scenarios.

CCS Concepts

• Information systems → Recommender systems.

Keywords

Proactive News Recommendation, User Openness, Large Language Model

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1 Introduction

Over the past decade, personalized recommendation systems have become integral to navigating the vast and diverse content landscape of online news platforms. By learning from users' historical interaction logs, conventional sequence-based models aim to predict future clicks by reflecting a user's past preferences. However, this approach often traps users in "filter bubbles [1, 46]," progressively narrowing their exposure to diverse information [40] and heightening the risks of ideological polarization [18, 19, 55].

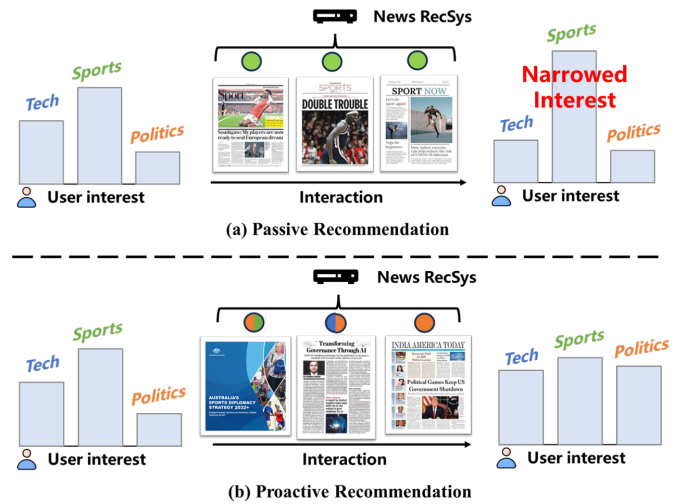


Figure 1: Illustration of Passive and Proactive News Recommendation. The color of the circle above each news item indicates the topic.

To mitigate the limitations of traditional models, a proactive recommendation paradigm is crucial. Instead of passively predicting user behavior, proactive recommendation seeks to actively shape a user's future interests by delivering a strategically curated sequence of news items. By progressively guiding users towards a predefined target news [53], proactive recommendation systems have the potential to broaden users' information horizons and contribute to a

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healthier content ecosystem [8]. This process can, therefore, help in shaping public opinion [10] while maintaining social stability.

Nevertheless, designing an effective proactive recommendation system presents three main challenges. First, it must model each user's receptiveness [4, 11, 56] to novel news content that diverges from their usual preferences. Second, it needs to track users' evolving interests [39] throughout the multi-round interaction cycle. Finally, the system must balance immediate relevance with long-term target guidance when selecting intermediary news items.

To address these challenges, we introduce **ONeRec**, a novel framework for Openness-aware and adaptive proactive **News Recommendation**. The core idea behind ONeRec is to guide a user's interest towards a selected target news item over multiple recommendation rounds. ONeRec adapts to the user's openness and current interests, progressively recommending news items that balance immediate relevance with long-term guidance. ONeRec integrates two personalized mechanisms to effectively guide users. The first is an openness coefficient, derived from the user's historical reading patterns. This coefficient quantifies the user's openness to novel news content [11, 43, 56] by evaluating the diversity of topics they engage with and their frequency of category transitions. It helps to balance immediate interest matching with the longer-term goal of guiding the user towards new, relevant news items. The second mechanism is the evolutionary coefficient, which adapts the user's interest profile dynamically [54] as they interact with recommended news items. This coefficient, derived from the dissimilarity between adjacent news items in the user's history, controls the rate at which the user's interests evolve, ensuring a personalized and adaptive recommendation flow. Furthermore, to facilitate offline evaluation among multiple rounds, we incorporate a Large Language Model (LLM) agent to simulate user feedback [2, 12, 32, 36].

The main contributions of this paper are summarized as follows:

- (1) To our best knowledge, this is the first effort to address the proactive news recommendation task. We introduce **ONeRec**, a novel framework towards Openness-aware and adaptive proactive **News Recommendation**, which gradually cultivates a user's interest in a predefined target news article.
- (2) In ONeRec, we propose two key personalization mechanisms: an openness coefficient to evaluate a user's novelty tolerance and tailor the recommendation process, and an evolutionary coefficient to dynamically track short-term user interest shifts over multiple recommendation rounds.
- (3) We conduct comprehensive experiments on the public MIND dataset, and the results demonstrate that our model consistently outperforms state-of-the-art baselines across a range of proactive recommendation metrics, confirming its effectiveness.

The rest of this paper is structured as follows. We briefly review related work in Section 2. In Section 3, we introduce the preliminaries and formally define the problem. In Section 4, we detail our methodology. In Section 5, we present the experimental settings and results. Finally, we summarize our work and suggest future research directions in Section 6.

2 Related Work

We review two main strands of research that are pertinent to our work: personalized news recommendation and proactive recommendation.

2.1 Personalized News Recommendation

Personalized news recommendation aims to match users with articles that align with their interests [51], a field that has seen significant evolution.

Traditional Methods. Early approaches were dominated by conventional models. Collaborative Filtering [20] leverages user-item interaction matrices but struggles in the news domain due to the rapid content turnover and resulting data sparsity. Content-Based methods analyze textual content to model user interests, but their reliance on surface-level features often fails to capture deeper semantic nuances [33, 42].

Deep Learning-Based Sequential Models. More recently, the field has shifted towards deep learning-based methods, particularly those that model the sequential nature of user behavior [6]. These models have become the standard for news recommendation, employing architectures like Recurrent Neural Networks and Transformers [21, 49] to capture dynamic user interests from their chronological click histories [14, 58]. SASRec [26, 47], for example, successfully applied the self-attention mechanism to model sequential patterns [45]. Others have explored enhancing these models with contrastive learning [41], such as ICLRec [9], to better understand latent user intentions [30]. Graph-based methods have also been extensively explored to capture high-order structural information and disentangle user preferences [22, 23]. More recently, researchers have focused on integrating richer contextual signals to refine recommendation precision. For instance, DIVAN [15] introduces a virality-aware network to capture the complex temporal dynamics of news consumption, while SJORS [17] leverages semantic reasoning to tailor recommendations for specific professional domains.

Despite their increasing sophistication in matching users with relevant content, all the above methods are fundamentally passive. By design, they are optimized to predict the next click based on a user's established preferences [9, 26]. This user-centric focus can inadvertently confine users within "filter bubbles" or "information cocoons [18, 48]" by creating self-reinforcing feedback loops. Crucially, these systems lack any mechanism to expand user interests or proactively guide them towards predefined topical goals.

2.2 Proactive Recommendation

Proactive recommendation is an emerging research field that aims to actively and strategically steer user interests over multiple interaction rounds [44]. This area can be broadly categorized into two main directions.

Preference Evolution Modeling. This line of research primarily investigates how user preferences evolve through interactions with a recommender system [7], often using simulation techniques [12, 37]. The goal is typically to optimize for long-term user engagement and satisfaction [13, 50], rather than focusing on myopic, next-click prediction.

User Preference Guiding. This research stream, with roots in dialogue and conversational systems, focuses on explicitly nudging users towards a predefined target item or topic. Its application in sequential recommendation is a more recent development. For instance, IRN [57] adapted the Transformer architecture by unmasking the target item to inject target awareness, but it operates on the strong assumption that users will passively accept all intermediary items. Addressing some of these limitations, IPG [5] a model-agnostic, post-processing framework to re-rank items by blending relevance with a guidance value; yet, its validation relied on synthetic data and a simplistic threshold-based click model, limiting its applicability to real-world scenarios. More recently, ITMPRec [35] introduced components like intention modeling [25] and LLM-based feedback simulation. Nevertheless, its approach of assigning the same popular targets to all users and its lack of an explicit mechanism to model a user’s receptiveness to novelty highlight remaining gaps in personalizing the guidance strategy.

In summary, while existing proactive methods have established the feasibility of user guidance, they often overlook the critical importance of personalizing the guidance strategy based on a user’s intrinsic traits, such as their openness to novelty. Our work addresses this gap by proposing a framework that explicitly models this characteristic and uses it to dynamically balance the trade-off between immediate relevance and long-term guidance.

3 Preliminaries

In this section, we formalize the task of proactive news recommendation. We begin by introducing the key notations used throughout this paper, followed by a formal problem definition that distinguishes proactive recommendation from conventional paradigms.

Let $\mathcal{U} = \{u_1, u_2, \dots\}$ denote the set of users and $\mathcal{I} = \{i_1, i_2, \dots\}$ be the set of news items. Each user $u \in \mathcal{U}$ is associated with a chronologically ordered historical interaction sequence, $H_u = (i_1, i_2, \dots, i_t)$, where $i_k \in \mathcal{I}$ is the k -th news item with which the user has interacted. Our objective is to model users and items in a shared d -dimensional latent space, where a user u is represented by an embedding $\mathbf{e}_u \in \mathbb{R}^d$ and an item i is represented by an embedding $\mathbf{e}_i \in \mathbb{R}^d$. The key notations are summarized in Table 1.

Table 1: Main notations used in this paper.

Symbol	Description
$\mathcal{U}, \mathcal{I}$	Sets of all users and news items.
u, i, j	User, intermediary item, and target item.
d	Dimensionality of the embedding space.
H_u	Historical click sequence of user u .
S_u	Embedding sequence for H_u .
$\mathbf{e}_u, \mathbf{e}_i \in \mathbb{R}^d$	User and item embedding vectors.
R	Total number of recommendation rounds.
r	Index of the current recommendation round.
$\mathbf{e}_u^r$	User embedding at the start of round r .
i_u^r	Intermediary item for user u in round r .
$f_u^r \in \{0, 1\}$	Simulated user feedback (1=accept, 0=reject).

3.1 Conventional News Recommendation

Conventional news recommendation, particularly sequence-based models, primarily aims to predict the next item a user is likely to click. Given a user’s historical interaction sequence $H_u = (i_1, \dots, i_t)$, the corresponding embedding sequence $S_u = (\mathbf{e}_1, \dots, \mathbf{e}_t)$ is fed into a sequence encoder to derive the user’s current interest representation $\mathbf{e}_u$:

$$\mathbf{e}_u = \text{Encoder}(S_u) \quad (1)$$

The system then recommends items that maximize a scoring function, which typically measures the similarity between the user’s interest and candidate items, such as the dot product:

$$\text{score}(u, i) = (\mathbf{e}_u)^T \mathbf{e}_i \quad (2)$$

This approach is inherently passive, as it exclusively models and caters to a user’s past preferences without an external guiding objective.

3.2 Proactive News Recommendation

In contrast, proactive news recommendation operates as a multi-round, objective-driven process. The system is tasked not only with matching a user’s current interests but also with actively steering them towards a predetermined target item $j \in \mathcal{I}$.

Formally, for a given user u with an initial interest representation $\mathbf{e}_u^0$, the goal is to generate a sequence of intermediary recommendations $\{i_u^1, i_u^2, \dots, i_u^R\}$ over R rounds. In each round r , the system recommends an item i_u^r . Based on simulated user feedback $f_u^r \in \{0, 1\}$, the user’s state is updated. If the user accepts the item ($f_u^r = 1$), their history is appended and their interest representation evolves:

$$H_u^{r+1} = H_u^r \oplus i_u^r \quad (3)$$

$$\mathbf{e}_u^{r+1} = \text{Update}(\mathbf{e}_u^r, \mathbf{e}_{i_u^r}) \quad (4)$$

where $\oplus$ denotes concatenation and the $\text{Update}(\cdot)$ function models the shift in the user’s interest based on the newly consumed item. If the item is rejected ($f_u^r = 0$), the user’s state may remain unchanged or be adjusted differently.

The fundamental objective of the proactive recommendation task is to guide the user’s interest such that their affinity for the target item j after r rounds is greater than their initial affinity. This can be expressed as the condition that the user-target similarity score should increase:

$$(\mathbf{e}_u^r)^T \mathbf{e}_j > (\mathbf{e}_u^0)^T \mathbf{e}_j \quad (5)$$

This paradigm requires the system to delicately balance immediate relevance (to secure user acceptance of intermediary items) with long-term guiding value (to effectively navigate the user towards the target).

4 Methodology

In this section, we present our ONeRec framework in detail. As illustrated in Figure 2, the architecture consists of five key components that work in tandem within a multi-round recommendation process. These components guide users towards a target news item through a combination of dynamic interest modeling and personalized interaction strategies.

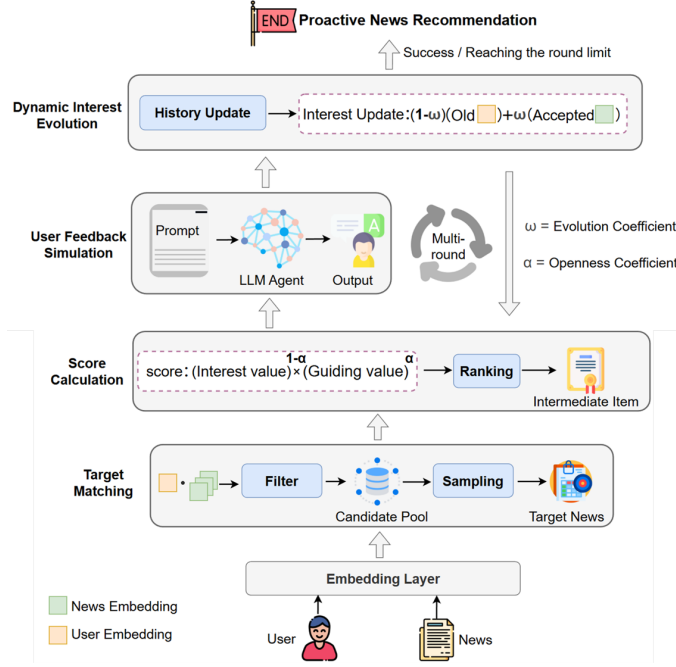


Figure 2: The overall framework of ONERec.

4.1 Embedding Layer

The foundation of our model lies in effective user and news representations.

4.1.1 User Embedding. We model a user's initial interest based on their historical reading sequence. Given the sequence of news embeddings $S_u = (\mathbf{e}_1, \mathbf{e}_2, \dots, \mathbf{e}_t)$ corresponding to a user's history H_u , we employ a sequence encoder to produce the initial user embedding $\mathbf{e}_u \in \mathbb{R}^d$:

$$\mathbf{e}_u = \text{Encoder}(S_u) \quad (6)$$

4.1.2 News Embedding. A news item's representation is a composite of its textual content and categorical information [3]. For a news item i with title t_i and a set of category words $\{w_{i,1}, \dots, w_{i,k}\}$, we first encode the title using a pre-trained sentence encoder to obtain a title embedding $\mathbf{e}_i^{\text{title}}$. We then embed the category words using an encoder like Word2Vec and average them to get a category embedding $\mathbf{e}_i^{\text{cat}}$. The final news embedding $\mathbf{e}_i \in \mathbb{R}^d$ is the concatenation of these two vectors:

$$\mathbf{e}_i^{\text{title}} = \text{SentenceEncoder}(t_i) \quad (7)$$

$$\mathbf{e}_i^{\text{cat}} = \frac{1}{k} \sum_{j=1}^k \text{WordEncoder}(w_{i,j}) \quad (8)$$

$$\mathbf{e}_i = \text{Concat}(\mathbf{e}_i^{\text{title}}, \mathbf{e}_i^{\text{cat}}) \quad (9)$$

4.2 Target Matching Module

To establish a meaningful guidance objective, first, we randomly sample a pool of candidate news items C_u from the entire corpus. We then estimate an "acceptance probability" for each item by computing the similarity between the user and item embeddings. This

pool is filtered to retain only items with an acceptance probability below 0.5, as items with higher probabilities are likely already aligned with the user's current interests. Finally, we randomly sample a single target item T_u from this filtered set to serve as the guidance objective. This process is formally expressed as:

$$T_u = \text{RandomSample}(\{i \in C_u \mid \sigma(\mathbf{e}_u^T \mathbf{e}_i) \leq 0.5\}) \quad (10)$$

4.3 Openness-Aware Score Calculation

At the core of ONERec is its scoring mechanism, which evaluates intermediary items by dynamically weighting their immediate relevance against their long-term guiding value, personalized by a user's receptiveness to novelty [43, 56].

4.3.1 Openness Coefficient (α). We quantify each user's openness to unfamiliar content with a coefficient α . This coefficient is derived from two metrics computed over the user's historical interactions H_u . This coefficient is computed once for each user before the recommendation process begins and remains constant throughout all guiding rounds, as it reflects a relatively stable personality trait over the short term.

- **Category Transition Frequency (F):** Measures how often a user switches between news categories. It is the number of times adjacent items in the history belong to different categories, normalized by sequence length.

$$F = \frac{1}{|H_u| - 1} \sum_{k=1}^{|H_u|-1} \mathbb{I}(c_{h_k} \neq c_{h_{k+1}}) \quad (11)$$

where $\mathbb{I}(\cdot)$ is the indicator function and c_{h_k} is the category of the k -th historical item.

- **Historical Content Diversity (D):** Measures the average dissimilarity among all items in a user's history, capturing the breadth of their interests.

$$D = 1 - \frac{2}{|H_u|(|H_u| - 1)} \sum_{1 \leq k < j \leq |H_u|} \text{cosine}(\mathbf{e}_{h_k}, \mathbf{e}_{h_j}) \quad (12)$$

The final openness coefficient α is a weighted combination of these two factors, anchored by a base value:

$$\alpha = \lambda_0 + \lambda_1(F + D) \quad (13)$$

where λ_0 and λ_1 are tunable hyperparameters.

To better clarify the calculation of α , consider a user with a short reading history $H_u = \{i_1, i_2, i_3, i_4\}$ consisting of four items with the categories [Politics, Politics, Finance, Sports].

First, for the *Category Transition Frequency (F)*, we observe the transitions between adjacent items: $i_1 \rightarrow i_2$ (Politics $\rightarrow$ Politics, no change), $i_2 \rightarrow i_3$ (Politics $\rightarrow$ Finance, change), and $i_3 \rightarrow i_4$ (Finance $\rightarrow$ Sports, change). This results in two category shifts out of three total transitions, yielding $F = \frac{0+1+1}{3} \approx 0.67$. A high F indicates a user's dynamic browsing habit.

Second, for the *Historical Content Diversity (D)*, we examine the semantic space. Although i_1 and i_2 have high cosine similarity (being in the same category), i_3 and i_4 are semantically distinct from the others. Consequently, the average pairwise cosine similarity across the set is low, resulting in a high value for D (as $D = 1 - \text{average similarity}$).

Finally, combining these factors, the resulting Openness Coefficient $\alpha = \lambda_0 + \lambda_1(F + D)$ will be relatively high. This accurately

characterizes the user as “open-minded,” prompting ONeRec to assign greater weight to the guiding value (g_{val}) in the subsequent scoring phase to actively steer them toward the target.

4.3.2 Composite Scoring. The comprehensive score for an intermediary item i is a product of its interest value and guiding value, with their relative importance controlled by α .

- **Interest Value (i_{val}):** The immediate relevance of item i to user u , calculated as the sigmoid-normalized dot product of their embeddings.

$$i_{val} = \sigma(\mathbf{e}_u^T \mathbf{e}_i) \quad (14)$$

- **Guiding Value (g_{val}):** The expected progress towards the target item j if the user consumes item i . Calculating the exact user interest update for every candidate item is computationally intensive. Therefore, we approximate the progress by assuming the user’s interest vector shifts linearly from its current position $\mathbf{e}_u$ towards the intermediary item’s vector $\mathbf{e}_i$ [5]. The vector $(\mathbf{e}_i - \mathbf{e}_u)$ represents this potential shift. We then project this shift vector onto the target item’s vector $\mathbf{e}_j$. This measures how much the interaction would move the user’s interest closer to the target. This value is normalized using the sigmoid function:

$$g_{val} = \sigma((\mathbf{e}_i - \mathbf{e}_u)^T \mathbf{e}_j) \quad (15)$$

The final score combines these two components, modulated by the user’s openness α :

$$\text{score}(u, i, j) = (i_{val} + \epsilon)^{1-\alpha} \cdot (g_{val} + \epsilon)^\alpha \quad (16)$$

where ϵ is a small positive constant to prevent the score from being zero. A higher α places more weight on the guiding value for open-minded users, while a lower α prioritizes interest matching for users with narrower tastes. ONeRec recommends the item with the highest score in each round.

4.4 User Feedback Simulation

To enable offline training and evaluation [24], we simulate user responses to recommended items using an LLM agent [12, 29, 32, 34, 36]. This process is formally represented as:

$$f_u^r = \text{LLM}(\mathcal{P}_u, H_u^r, \text{Names}(i_u^r)) \quad (17)$$

where $f_u^r \in \{0, 1\}$ is the binary feedback (1 for acceptance, 0 for rejection). The prompt $\mathcal{P}_u$ contains user profile information, H_u^r is their current historical sequence, and $\text{Names}(i_u^r)$ is the title of the candidate news item. If the user rejects an item, the system adaptively reduces the weight of the guiding value in the subsequent round to propose an item that is easier to accept.

4.5 Dynamic Interest Evolution

Upon receiving user feedback, the model updates the user’s state.

4.5.1 Evolution Coefficient (ω). We introduce a personalized evolution coefficient ω to control the rate at which a user’s interest representation is updated. This coefficient reflects the user’s intrinsic sensitivity to new information and is derived from the average dissimilarity between adjacent items in their historical sequence.

The raw value is then normalized and clipped to a predefined range $[a, b]$ to ensure stability:

$$\omega_{\text{raw}} = \frac{1}{|H_u| - 1} \sum_{k=1}^{|H_u|-1} (1 - \text{cosine}(\mathbf{e}_{h_k}, \mathbf{e}_{h_{k+1}})) \quad (18)$$

$$\omega = \text{Clip}(\text{Norm}(\omega_{\text{raw}}), [a, b]) \quad (19)$$

4.5.2 Interest and History Update. If the user accepts the recommended item ($f_u^r = 1$), the user’s embedding is updated via a linear interpolation between their old embedding and the embedding of the accepted item, controlled by ω :

$$\mathbf{e}_u^{r+1} = (1 - \omega)\mathbf{e}_u^r + \omega\mathbf{e}_{i_u^r} \quad (20)$$

Simultaneously, the user’s history is appended with the newly accepted item:

$$H_u^{r+1} = \text{Concat}(H_u^r, \text{Names}(i_u^r)) \quad (21)$$

If the item is rejected ($f_u^r = 0$), the user’s embedding and history remain unchanged for the next round. The process then repeats for the next guidance round.

5 Experiments

In this section, we conduct a series of experiments to evaluate the performance of our model, ONeRec. We aim to answer the following research questions (RQs):

- **RQ1:** How does ONeRec perform in comparison to state-of-the-art sequential and proactive recommendation methods?
- **RQ2:** What is the effectiveness of ONeRec’s core components, such as the openness and evolution coefficients?
- **RQ3:** How does ONeRec’s proactive guidance mechanism work in a practical, user-specific scenario?

5.1 Experiment Settings

5.1.1 Dataset and Preprocessing. Our experiments are conducted on the widely-used **MIND (Microsoft News Dataset)** [52], a large-scale public benchmark for news recommendation research. We use the MIND-small version, which contains anonymized interaction logs from 50,000 users. The dataset includes `news.tsv`, which provides article information such as title and category, and `behaviors.tsv`, which contains users’ chronological click histories and impression data.

For preprocessing, we follow a standard procedure. We process the `behaviors.tsv` file to construct a comprehensive historical interaction sequence for each user. To ensure data quality and focus on users with established preferences, we filter out users with fewer than five interactions. For computational efficiency and to emphasize recent interests, we truncate each user’s history to their last 30 clicked articles. From `news.tsv`, we retain the essential fields for our model: news ID, title, and category.

5.1.2 Evaluation Metrics. To provide a comprehensive evaluation of proactive recommendation performance, we adopt three standard metrics, evaluated at different stages of the process, specifically at rounds $k \in \{5, 10, 15, 20\}$.

- **Hit Rate (Hit@k):** This metric measures the immediate relevance and user acceptance of the intermediary items. It

Table 2: The overall experiment results on MIND dataset. The bold is the best performance and the underline is the second-best.

Methods	Hit@5	IOI@5	IOR@5	Hit@10	IOI@10	IOR@10	Hit@15	IOI@15	IOR@15	Hit@20	IOI@20	IOR@20
ICLRec	0.4184	0.0021	-3.0020	0.4300	0.0035	-3.0020	0.4328	0.0043	1.1720	0.4391	0.0049	-87.9960
SASRec	0.4212	-0.0030	-158.8780	<u>0.4298</u>	-0.0022	-203.4040	<u>0.4170</u>	-0.0072	-35.1050	<u>0.4239</u>	0.0061	150.7950
SASRec-IPG	0.3781	0.2454	1084.3470	0.3781	0.2454	1084.3470	0.3804	0.6736	3276.0600	0.3805	0.8694	3083.5413
ITMPRec	0.4065	<u>0.3339</u>	<u>1747.3200</u>	0.3968	<u>0.6566</u>	<u>2925.9857</u>	0.4062	<u>0.7762</u>	<u>3829.1980</u>	0.4001	<u>0.8799</u>	<u>4752.3150</u>
ONeRec	0.4152	0.4128	2471.1980	0.4186	0.6573	3539.6175	0.4184	0.9863	5402.6475	0.4181	1.2665	6965.2288

is calculated as the ratio of accepted recommendations to the total recommendations made up to round k . Formally:

$$\text{Hit}@k = \frac{1}{|\mathcal{U}| \cdot k} \sum_{u \in \mathcal{U}} \sum_{r=1}^k f_u^r \quad (22)$$

where $f_u^r \in \{0, 1\}$ is the simulated feedback for user u in round r .

- **Increase of Interest (IOI@k):** This metric quantifies the effectiveness of the guidance process by measuring the change in a user's affinity for the target item j . It is defined as the average increase in the user-target similarity score after k rounds.

$$\text{IOI}@k = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \left((e_u^k)^T e_j - (e_u^0)^T e_j \right) \quad (23)$$

where e_u^0 and e_u^k are the user embeddings at the initial state and after round k , respectively.

- **Increase of Ranking (IOR@k):** This metric measures the improvement in the target item's relevance from the user's perspective. It is calculated as the average improvement in the rank of the target item j within a candidate pool.

$$\text{IOR}@k = \frac{1}{|\mathcal{U}|} \sum_{u \in \mathcal{U}} \left(\text{Rank}(e_j | e_u^0) - \text{Rank}(e_j | e_u^k) \right) \quad (24)$$

where $\text{Rank}(e_j | e_u^*)$ denotes the rank of the target item j based on its similarity to the user embedding e_u^* at a given round.

5.1.3 Baselines. We compare ONeRec against two categories of strong baseline models:

- **Sequential Recommendation Models:** These models are designed to predict the next item a user will interact with but lack an explicit guidance mechanism.
 - **SASRec** [26]: A classic self-attention-based model that captures sequential patterns in user behavior.
 - **ICLRec** [9]: An intent-aware model that uses contrastive learning to better understand user intentions in sequential recommendation.
- **Proactive Recommendation Models:** These models are explicitly designed for target-oriented guidance.
 - **SASRec-IPG** [5]: This baseline combines the SASRec model with IPG, a post-processing framework that re-ranks recommendations to balance relevance and guiding value.
 - **ITMPRec** [35]: A proactive method that incorporates user intention modeling and an arousal coefficient to guide users towards a target.

5.1.4 Implementation Details. We implement our model and all baselines using PyTorch. User and news embeddings are generated as described in Section 4. Specifically, news titles are encoded using the all-MiniLM-L6-v2 SentenceTransformer model, and categories are embedded using a Word2Vec model. User embeddings are produced by a sequence encoder trained on historical interactions. To evaluate performance at different interaction depths, we conduct experiments where the maximum number of proactive recommendation rounds is set to $k \in \{5, 10, 15, 20\}$, and report results at each of these intervals. In line with our methodology, the user's openness coefficient is computed once at the beginning of the process, while the target item for each user is selected from a candidate pool of items with a predicted acceptance probability below 50%. User feedback in all experiments is simulated using an LLM agent to ensure a realistic and dynamic interaction environment.

5.2 RQ1: Overall Performance Comparison

The main experimental results are presented in Table 2. From the results, we can draw several key observations.

First, a clear distinction exists between proactive and conventional sequential models. The sequential baselines, SASRec [26] and ICLRec [9], achieve high Hit Rates because they are optimized to recommend items that align with users' past preferences. However, they exhibit negligible or even negative performance on the IOI and IOR metrics. This is expected, as these models lack any mechanism for guidance; their recommendations may drift away from a predefined target, confirming that high immediate relevance does not translate to effective long-term guidance.

Second, among the models designed for proactive recommendation, our ONeRec framework consistently and significantly outperforms all baselines on the core guidance metrics, IOI and IOR. Specifically, we observe that the baseline SASRec-IPG, which employs a post-processing re-ranking strategy to inject guidance, manages to improve IOI and IOR compared to the passive SASRec. However, this comes at a significant cost to user engagement, as evidenced by its Hit Rate being the lowest among all methods (e.g., 0.3805 at round 20). This suggests that a rigid re-ranking strategy tends to sacrifice user acceptance for guidance. In contrast, ONeRec achieves the highest scores for IOI and IOR across all evaluation intervals ($k = 5, 10, 15, 20$) while maintaining a competitive Hit Rate. The performance gap widens as the number of rounds increases, demonstrating the superiority of our model in sustained, multi-round guidance scenarios. For instance, at 20 rounds, ONeRec achieves an IOI@20 of 1.2665 and an IOR@20 of 6965.2288, substantially surpassing the next-best proactive model, ITMPRec [35].

The superior performance of ONeRec can be attributed to its unique design. The **openness-aware scoring mechanism** allows

for personalized guidance—making bolder, guidance-focused recommendations for users identified as open to novelty, while adopting a more cautious, interest-focused approach for others. This tailored strategy creates a more efficient path towards the target. Furthermore, the **dynamic interest evolution module**, controlled by a personalized coefficient, enables the model to adapt realistically to user feedback, making the multi-round updates more effective. Finally, ONeRec maintains a competitive Hit Rate, comparable to other proactive models and only slightly lower than purely sequential ones. This indicates a well-calibrated balance between achieving long-term guidance objectives and maintaining user engagement through relevant intermediary recommendations.

5.3 RQ2: Ablation Study

To rigorously evaluate the individual contributions of the core components within our ONeRec framework, we conduct a comprehensive ablation study. The objective is to understand the impact of removing or simplifying each key mechanism on the model’s overall performance, particularly its guidance capabilities. We create four variants of the full ONeRec model and compare their performance against the complete version after 20 recommendation rounds on the MIND dataset, using Hit@20, IOI@20, and IOR@20 as our primary evaluation metrics.

The variants are systematically designed as follows:

- **ONeRec w/o Openness:** This variant ablates the personalized openness coefficient (α). Instead of computing a unique α based on user history, we assign a fixed, non-personalized hyperparameter ($\alpha = 0.5$) for all users. This configuration gives equal weight to interest matching and guiding value, irrespective of individual user traits, thereby testing the importance of a personalized guidance strategy.
- **ONeRec w/o Evolution:** This variant removes the personalized evolution coefficient (ω). The personalized ω , which dictates the rate of user interest updates, is replaced with a small, fixed learning rate (e.g., $\omega = 0.1$) for all users. This experiment is designed to isolate the contribution of a dynamic, user-specific update mechanism versus a static one.
- **ONeRec w/o Guidance:** In this variant, the core guidance mechanism is removed from the scoring function. The composite score is based solely on the interest value (i_{val}), completely ignoring the guiding value (g_{val}). This effectively transforms the model into a passive recommender operating within a proactive loop, aiming to directly validate the necessity of the explicit guidance component.
- **ONeRec w/o Pre-Matching:** This variant eliminates the target pre-matching module. Instead of filtering the candidate pool to select targets with an acceptance probability below 50%, we select target items completely at random from the entire news corpus. This experiment will demonstrate the importance of initializing the guidance process with a suitable target.

The results of our ablation study are presented in Table 3. Our analysis of these results leads to several key insights.

First, removing the personalized openness coefficient (**ONeRec w/o Openness**) causes a significant degradation in guidance performance, with IOI@20 dropping from 1.2665 to 0.9341 and IOR@20

Table 3: Ablation study results on the MIND dataset, evaluated at 20 recommendation rounds. Performance drops in all variants confirm the effectiveness of each component.

Method	Hit@20	IOI@20	IOR@20
ONeRec (Full Model)	0.4181	1.2665	6965.2288
ONeRec w/o Openness	0.4095	0.9341	5102.7314
ONeRec w/o Evolution	0.4132	1.0872	6251.4890
ONeRec w/o Guidance	0.4255	0.0053	142.3017
ONeRec w/o Pre-Matching	0.3978	0.8116	4588.1923

decreasing by over 26%. This confirms our hypothesis that a uniform guidance strategy is suboptimal. For users with narrow interests, a fixed strategy may be too aggressive, leading to rejections, while for users open to novelty, it may be too conservative, resulting in an inefficient guidance path.

Second, replacing the personalized evolution coefficient with a fixed learning rate (**ONeRec w/o Evolution**) also leads to a clear decline in IOI and IOR scores. A fixed update rate fails to capture the nuances of how different users internalize new information, leading to less accurate user representations over the multi-round process and consequently affecting the efficacy of intermediate item selection.

Third, the most dramatic performance change is observed in the **ONeRec w/o Guidance** variant. As anticipated, the guidance-specific metrics collapse, with IOI@20 and IOR@20 plummeting to near-zero values (0.0053 and 142.3017, respectively), which is analogous to the performance of the sequential baselines. Conversely, its Hit@20 score (0.4255) is the highest among all variants, as it exclusively recommends items that match users’ existing preferences. This result strongly validates that the explicit guiding component is indispensable for achieving the objectives of proactive recommendation.

Finally, the **ONeRec w/o Pre-Matching** variant shows a substantial drop in guidance effectiveness. Randomly selecting targets can lead to a failure mode: choosing a target the user already likes, which makes the guidance trivial (near-zero IOI). This underscores the criticality of the pre-matching module for defining a meaningful guidance problem for each user.

In summary, the ablation results collectively verify that each of our components—personalized openness, dynamic evolution, a dedicated guiding score, and target matching—is a crucial and integral part of the ONeRec framework’s success.

5.4 RQ3: Case Study

To illustrate ONeRec’s adaptive guidance, we analyze a specific user whose history is dominated by Politics and Finance, resulting in a low Openness Coefficient (α). This indicates a preference for familiarity over novelty. The objective is to guide this user toward a target in the Sports category. Table 4 details the recommendation paths generated by ONeRec and the passive baseline SASRec [26] over ten rounds.

ONeRec adopts a “bridging” strategy, initially recommending articles that connect the user’s existing interests with the target domain (e.g., political or financial aspects of sports). This respects

Table 4: A case study illustrating the recommendation paths of ONeRec and a passive baseline (SASRec) over ten rounds.

Target News Article in Sports: <i>Data-Driven Strategies in Modern Sports</i>	
User's Recent History (Categories): Politics → Finance → Politics → Health → Finance	
ONeRec's Recommendation Path	Passive Baseline's (SASRec) Path
1. <i>Pol. Influence on Sports Funding</i> (Politics/Finance/Sports)	1. <i>Recent Election Updates</i> (Politics)
↓	↓
2. <i>Fin. Impacts of Olympic Events</i> (Finance/Sports)	2. <i>Stock Market Trends</i> (Finance)
↓	↓
3. <i>Health Policies in Pro Athletics</i> (Health/Sports)	3. <i>Global Health Initiatives</i> (Health)
↓	↓
4. <i>Econ. Aspects of Team Management</i> (Finance/Sports)	4. <i>Pol. Debates on Econ. Reform</i> (Politics/Finance)
↓	↓
5. <i>Athlete Wellness Programs</i> (Health/Sports)	5. <i>Fin. News Roundup</i> (Finance)
↓	↓
6. <i>Sports League Governance</i> (Politics/Sports)	6. <i>Health Policy Analysis</i> (Health/Politics)
↓	↓
7. <i>Investing in Sports Franchises</i> (Finance/Sports)	7. <i>Pol. Scandals Update</i> (Politics)
↓	↓
8. <i>Physical Fitness in Competitive Sports</i> (Health/Sports)	8. <i>Econ. Indicators Report</i> (Finance)
↓	↓
9. <i>Strategic Planning in Pro Teams</i> (Sports)	9. <i>Intl. Health Challenges</i> (Health)
↓	↓
10. <i>Modern Coaching Techniques</i> (Sports)	10. <i>Pol. Campaign Strategies</i> (Politics)
Target Category Reached ✓	Fails to Reach Target ✗

the user's low α by balancing immediate relevance with guidance. As the user accepts these items, the Evolutionary Coefficient (ω) dynamically updates their interest profile, allowing the model to progressively emphasize the guiding value. Consequently, ONeRec successfully transitions the user to pure Sports content, raising the user-target similarity score from 0.32 to 0.78.

In contrast, SASRec remains confined to the user's historical preferences, cycling through Politics and Finance without making progress toward the target. This comparison highlights ONeRec's superiority in mitigating filter bubbles: by integrating openness-aware scoring and dynamic interest evolution, it effectively broadens the user's information horizon in a personalized, user-centric manner.

6 Conclusion

In this paper, we introduced **ONeRec**, a proactive news recommendation framework designed to mitigate filter bubbles by strategically guiding users toward new topics. We proposed two personalized mechanisms—the *Openness Coefficient* and the *Evolutionary Coefficient*—to model user receptiveness and dynamic interest shifts. Extensive experiments on the MIND dataset demonstrate that ONeRec significantly outperforms baselines in achieving long-term guidance objectives while maintaining user engagement. Future work may explore the integration of causal inference [16, 38] to further enhance the model's explainability and robustness.

Crucially, we must emphasize the limitations regarding evaluation and ethics. First, our validation relies on offline LLM simulations; thus, the results may not fully reflect complex real-world behaviors, necessitating future human-in-the-loop studies. Second, ONeRec represents a methodological contribution rather than a

deployment-ready system. The capability to steer interests carries inherent risks of misuse, such as *opinion manipulation* or creating artificial “filter bubbles.” Therefore, any real-world application must be governed by strict transparency and user consent protocols. We provide a comprehensive discussion on these ethical implications and abuse prevention strategies in the **Ethics Statement** (see Appendix A.1).

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A Appendix

A.1 Ethics Statement

We acknowledge that the proactive news recommendation paradigm proposed in ONeRec—which actively steers user interests toward specific target content—raises important ethical considerations regarding user autonomy, potential manipulation, and transparency. While our primary research motivation is to mitigate the “filter bubble” effect and broaden users’ information horizons, we recognize the dual-use nature of such technology. To address the concerns regarding potential misuse for opinion manipulation or guiding users against their best interests, we outline the following ethical principles and safeguards integral to the proposed framework:

Respecting User Autonomy via Design.

Unlike coercive steering mechanisms, ONeRec is explicitly designed to respect user autonomy through the *Openness Coefficient* (α). This mechanism acts as an inherent safeguard: it quantifies a user’s natural willingness to explore new topics based on their historical behavior. For users who prefer familiarity (low openness), the model automatically reduces the weight of the guiding value, thereby preventing “forced” steering or aggressive content pushing. The system is designed to nudge only when the user exhibits a propensity for exploration, ensuring that the guidance aligns with the user’s psychological receptiveness.

Transparency and Informed Consent.

In any real-world deployment or online A/B testing of ONeRec,

strict transparency mechanisms must be implemented. Users should be:

- Explicitly informed that the recommendation system may proactively suggest diverse content to broaden their interests (Informed Consent).
- Provided with interface-level explanations (e.g., labels such as “To broaden your view”) when a recommendation is made primarily for guidance purposes rather than immediate relevance (Explainability).
- Granted the ability to opt-out of proactive guidance features or reset their interest evolution profile (User Control).

Prevention of Misuse and Content Safety.

To prevent the system from being used to amplify harmful propaganda or misinformation, the selection of “target items” must be rigorously controlled. The candidate pool for target news must undergo strict safety filtering. Advanced techniques for misinformation identification and fake news detection [31] should be integrated to ensure compliance with community guidelines and legal standards. We strictly condemn the use of proactive recommendation for political manipulation or the promotion of extremist content. The proposed method is intended solely for benevolent purposes, such as educational guidance or diversifying news consumption to foster a healthier public discourse.

Data Privacy.

Our experiments were conducted on the public MIND dataset with anonymized user IDs. No personally identifiable information (PII) was processed or stored. Future applications of this framework must adhere to relevant data protection regulations (e.g., GDPR) [28] and adopt clear data licensing frameworks [27] to ensure user privacy and data reusability are properly managed during the interest modeling process.

A.2 Complexity Analysis

To demonstrate the feasibility of ONeRec in production environments [45], we analyze the time complexity of its offline preprocessing and online inference phases. Let $|U|$ and $|I|$ denote the total number of users and news items, respectively. Let d be the embedding dimension, L be the maximum length of a user’s interaction history (e.g., $L = 30$), and M be the number of candidate items to be ranked in each round.

Offline Preprocessing Cost.

The initialization of the Openness Coefficient (α) and the Evolution Coefficient (ω) is performed offline or asynchronously.

- Computing α involves calculating the Category Transition Frequency ($O(L)$) and Historical Content Diversity. The latter requires pairwise cosine similarity calculations within the user history, leading to a complexity of $O(L^2 \cdot d)$. Since L is a small constant (typically $L \ll |I|$), this overhead is negligible.
- The Target Matching process filters the candidate pool based on dot-product similarity, with a complexity of $O(|C_u| \cdot d)$. As this is a one-time operation per session setup, it does not impact real-time serving latency.

Online Inference and Scalability.

During the online serving phase, the system needs to score M candidate items. The complexity for processing a single item i in ONeRec involves:

- (1) *Interest Value Calculation*: A standard dot product between user and item embeddings: $O(d)$.
- (2) *Guiding Value Calculation*: This requires computing $\sigma((e_i - e_u)^T e_j)$. This can be expanded as $\sigma(e_i^T e_j - e_u^T e_j)$. Since the term $e_u^T e_j$ is constant for all candidates in a given round, it can be pre-calculated. Thus, the marginal cost is dominated by $e_i^T e_j$, which is another dot product: $O(d)$.

Consequently, the total time complexity for ranking M candidates is $O(M \cdot d)$. This shares the same linear complexity order as standard inner-product-based retrieval models (e.g., SASRec or Matrix

Factorization). Furthermore, the update of user embeddings and history after feedback is an $O(d)$ operation.

Production Feasibility.

The space complexity is determined by the storage of embeddings $O((|U| + |I|) \cdot d)$, which is standard for latent factor models. Given that the online inference relies solely on efficient vector operations (dot products and element-wise additions), ONeRec can be easily accelerated using standard serving libraries (e.g., FAISS) or GPU inference. Therefore, ONeRec introduces minimal computational overhead compared to passive baselines and is highly scalable for real-world production systems.